

Discussion 6 Worksheet

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Problem 1: Rank and Solution Analysis

Let $\mathbf{A}$ be a 3×3 matrix and $\mathbf{b} \in \mathbb{R}^3$. Consider the linear system $\mathbf{Ax} = \mathbf{b}$. Given

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 6 \\ 12 \\ 3 \end{bmatrix}$$

Analyze the following questions based on the given system:

1. Compute the rank of $\mathbf{A}$.
2. Compute the rank of the augmented matrix $[\mathbf{A}|\mathbf{b}]$.
3. Compare the ranks of $\mathbf{A}$ and $[\mathbf{A}|\mathbf{b}]$. What does this result imply?

Problem 2: Rank and Solution Analysis

Let $\mathbf{A}$ be a 3×4 matrix and $\mathbf{b} \in \mathbb{R}^3$. Consider the linear system $\mathbf{Ax} = \mathbf{b}$. Given

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 1 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 7 \end{bmatrix}$$

Answer the following:

1. Compute the rank of $\mathbf{A}$.
2. Compute the rank of the augmented matrix $[\mathbf{A}|\mathbf{b}]$.
3. Compare the ranks of $\mathbf{A}$ and $[\mathbf{A}|\mathbf{b}]$. What does this result imply?

Problem 3: Transpose and Inverse

Prove that the inverse of $\mathbf{A}$ is $\mathbf{A}^\top$ for any value of θ if $\mathbf{A}$ is

$$\mathbf{A} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}.$$

Hint: $\cos^2(\theta) + \sin^2(\theta) = 1$ for any θ .

Problem 4: Determinant

Compute determinant for $\mathbf{A}$ for each case:

1. $\mathbf{A} = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix}$

2. $\mathbf{A} = \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix}$

3. $\mathbf{A} = \begin{bmatrix} a & 0 & 0 & 0 \\ b & c & 0 & 0 \\ d & e & f & 0 \\ g & h & i & j \end{bmatrix}$

4. Generalize the determinant of $n \times n$ lower triangular matrix $\mathbf{A}$.